

FINAL TERM EXAMINATION

MTH302- Business Mathematics & Statistics (Session - 1)

Question No: 1 (Marks: 1) - Please choose one

The c2 distribution is -----

- ▶ Positively skewed
- ▶ **Negatively skewed**
- ▶ Symmetrical
- ▶ None of these

Question No: 2 (Marks: 1) - Please choose one

In the regression line $Y = a + bX$, a represents

- ▶ **Intercept of the line**
- ▶ Slope of the line
- ▶ Slope of the curve
- ▶ None of the above

Question No: 3 (Marks: 1) - Please choose one

In the formula, $POISSON = \frac{e^{-\lambda} \lambda^x}{x!}$, e is a constant whose value is

- ▶ **2.71828**
- ▶ 3.14157
- ▶ 22 / 7
- ▶ 9.325

Question No: 4 (Marks: 1) - Please choose one

Which distribution is most commonly used for the continuous data?

- ▶ **Normal Distribution**
- ▶ Binomial Distribution
- ▶ Poisson Distribution
- ▶ Negative Binomial Distribution

Question No: 5 (Marks: 1) - Please choose one

For any event if the probability of success is x, then the probability of failure is

- ▶ also x
- ▶ $1 + x$
- ▶ **$1 - x$**

- ▶ $x - 1$

Question No: 6 (Marks: 1) - Please choose one

The standard deviation of any sampling distribution is called

- ▶ sampling error
- ▶ type –I error
- ▶ **standard error**
- ▶ non sampling error

Question No: 7 (Marks: 1) - Please choose one

For graphing univariate data we use

- ▶ Pie charts, Bar charts .
- ▶ Pareto diagrams.
- ▶ Side by side chart .
- ▶ **Both (1) & (2)**

Question No: 8 (Marks: 1) - Please choose one

Interest paid (earned) on only the original principal borrowed (lent) is often referred to as

- ▶ **simple interest**
- ▶ present value
- ▶ future value
- ▶ compound interest

Question No: 9 (Marks: 1) - Please choose one

The sales of a company increases from Rs.100,000 last month to Rs.120,000 this month.
The percentage change in profit for the month?

- ▶ **20%**
- ▶ 40%
- ▶ 10%
- ▶ 60%

Question No: 10 (Marks: 1) - Please choose one

Amount of discount is obtained as

- ▶ **Percentage of Discount x List Price**
- ▶ Percentage of Discount / List Price
- ▶ Percentage of Discount - List Price
- ▶ **None of these**

Question No: 11 (Marks: 1) - Please choose one

If a and b are any two integers such that $a < b$ then $a - b$ is

- ▶ positive
- ▶ **negative**
- ▶ cannot be told
- ▶ zero

Question No: 12 (Marks: 1) - Please choose one

If the salary of an employee is 10,000 and his allowances are 5,000 then what is the taxable income of the employee?

- ▶ 5,000
- ▶ 10,000
- ▶ **Zero**
- ▶ 15,000

Question No: 13 (Marks: 1) - Please choose one

Compound interest is always:

- ▶ **Higher than simple interest**
- ▶ Smaller than simple interest
- ▶ Less than annuity
- ▶ Equals to simple interest

Question No: 14 (Marks: 1) - Please choose one

The RSQ of any data is

- ▶ Square root of correlation coefficient.
- ▶ **Square of correlation coefficient.**
- ▶ Correlation coefficient.
- ▶ Slope of the data.

Question No: 15 (Marks: 1) - Please choose one

12% of the electric bulbs manufactured by a company are defective. What is the probability that a bulb selected will not be defective?

- ▶ 90 %
- ▶ **88 %**
- ▶ 80 %
- ▶ 85 %

Question No: 16 (Marks: 1) - Please choose one

If there is an indirect relationship between rainfall & yield of crops then

- ▶ **Yield is higher if rainfall is less**
- ▶ Yield is lower if rainfall is less
- ▶ Yield is higher if rainfall is higher

- Yield is higher if rainfall is nothing

Question No: 17 (Marks: 1) - Please choose one

If the estimating equation is $Y = a - bX$, which of the following is true

- The y intercept is b
- Slope of line is positive
- **There is inverse relationship**
- There is direct relationship

Question No: 18 (Marks: 1) - Please choose one

How many ways can 4 prizes be given away to 3 boys, if each boy is eligible for all the prizes?

- 256
- 12
- **81**
- None of these

<http://www.bestsamplequestions.com/psat-sample-questions/psat-sample-math-questions/psat-sample-multiple-choice-questions/psat-sample-multiple-choice-questions-15.html>

Question No: 19 (Marks: 1) - Please choose one

How many ways can 10 letters be posted in 5 post boxes, if each of the post boxes can take more than 10 letters?

- **5^{10}**
- 10^5
- $10P5$
- $10C5$

Question No: 20 (Marks: 1) - Please choose one

Equation of line having slope 0 and passing through the point A (0, 0) is

- $X=0$
- $Y=X$
- **$Y=0$**
- $X-1=0$

Ref $y - y_1 = m(x - x_1)$ $m = \text{slope}$ A(0,0) $x_1=0$ $y_1=0$

Question No: 21 (Marks: 2)

Define the Null Hypothesis.

ANS:

Null hypothesis is a scenario which explain a given set of data. It is tested to determine whether data provides sufficient reasons to pursue some alternative hypothesis. It is a hypothesis that states there is no difference between 2 or more sets of data.

Question No: 22 (Marks: 2)

A coin can be tossed in 3 ways. A die can be thrown in 6 ways. A coin and a die together can be thrown in ... **3 X 6 = 18**.....ways.

Question No: 23 (Marks: 2)

Find harmonic mean (HM) of 10,12,14,17.

ANS:

Harmonic mean, HM = $n / (1/x_1 + 1/x_2 + 1/x_3 + 1/x_4 + \dots + 1/x_n)$

N= 4

$X_1 = 10$

$X_2 = 12$

$X_3 = 14$

$X_4 = 17$

$1/x_1 + 1/x_2 + 1/x_3 + 1/x_4 = 1/10 + 1/12 + 1/14 + 1/17$

$1/10 = 0.1$

$1/12 = 0.08333$

$1/14 = 0.07142$

$1/17 = 0.05882$

Sum = 0.31357

Harmonic mean = $4 / 0.31357 = 12.756$

Question No: 24 (Marks: 3)

Find proportion of bags which have weight in excess of 505 g.

Mean = 500.

StDev = 2.5 g

(For $z=2.00$ probability is 0.4772)

Solution:

= .5 - 0.4772

= 0.0228

Question No: 25 (Marks: 3)

Given for a frequency distribution mode = 18, mean = 21. Calculate median

. Using these values comment on skewness of distribution.

ANS: We know that,

Mean-mode = $3(\text{mean} - \text{median})$

Median = $(2 * \text{mean} + \text{mode}) / 3$

Since,

Mean = 21

Mode = 18

So,

Median = $(2 * 21 + 18) / 3$

$= (42 + 18) / 3 = 60 / 3 = 10$

The distribution is Moderately skewed and unimodal distribution.

OR

Solution:

We know that,

Mean-mode = 3(mean - median)

Put the values in above equation:

$$21 - 18 = 3(21 - \text{Median})$$

$$3 = 63 - 3\text{median}$$

$$3\text{median} = 63 - 3$$

$$3\text{median} = 60$$

$$\text{Median} = 60/3$$

$$\text{Median} = 20$$

The distribution is moderately skewed and unimodal distribution

Question No: 26 (Marks: 3)

How many different ways can you select 2 letters from the set of letters: X, Y, and Z?
(Hint: In this problem, order is NOT important; i.e., XY is considered the same selection as YX.)

ANS:

Total number of letters = 3

Letters taken at a time = 2

Number of ways in which 2 letters can be selected out of 3 = ${}^3C_2 = 3! / 2!(3-2)!$

$$= 3 \times 2 \times 1 / 2$$

$$= 3$$

Question No: 27 (Marks: 5)

How many possible permutations can be formed from the word MATHEMATICS?

ANS:

Total number of alphabets in mathematics = 11

$$M = 2$$

$$A = 2$$

$$T = 2$$

$$H = 1$$

$$E = 1$$

$$I = 1$$

$$C = 1$$

$$S = 1$$

$$\text{Permutations} = 11! / 2! * 2! * 2! * 1! * 1! * 1! * 1! * 1! \\ = 11! / 8 = 4,989,600$$

Question No: 28 (Marks: 5)

Find the standard deviation of 4, 9, 11, 12, 17, 5, 8, 12, 14

ANS:

Standard deviation for sample = $\sqrt{\sum(X - \bar{X})^2 / n - 1}$

$$\bar{X} = \text{mean} = (4 + 9 + 11 + 12 + 17 + 5 + 8 + 12 + 14) / 9 = 92 / 9 = 10.22$$

$$n = 9$$

$$\text{so, } n - 1 = 9 - 1 = 8$$

X	X - $\bar{X}$	(X - $\bar{X}$)²
4	-6.22	38.69
9	-1.22	1.49
11	0.78	0.61
12	1.78	3.17
17	6.78	45.97
5	-5.22	27.25
8	-2.22	4.93
12	1.78	3.17
14	3.78	14.29

$$\sum(X - \bar{X})^2 = 139.57$$

$$\text{Standard deviation} = \sqrt{\frac{139.57}{8}} = 4.18$$

Question No: 29 (Marks: 5)

$$\mu = 30, \sigma = 5$$

In a normal distribution what proportion of cases will fall between 20 and 35?

Question No: 30 (Marks: 10)

Form a regression line from the data below.

Departments	X	Y
1	26	81
2	42	95
3	16	53
14	55	43
5	92	18
6	28	63
7	13	60

$$\text{ANS: Slope, } b = [n * \sum(X * Y) - \sum X * \sum Y] / [n * \sum X^2 - (\sum X)^2]$$

$$\text{intercept, } a = (\text{sum}Y - b * \text{sum}X) / n$$

$$n = 7$$

Departments	X	Y	X ²	X*Y
1	26	81	676	2106
2	42	95	1764	3990
3	16	53	256	848
4	55	43	3025	2365
5	92	18	8464	1656
6	28	63	784	1764
7	13	60	169	780

$$\text{sum}X = 272$$

$$\text{sum}Y = 413$$

$$\text{sum}X^2 = 15138$$

$$\text{sum}X * Y = 13509$$

$$\text{slope, } b = [7(13509) - 272 * 413] / [7(15138) - (272)^2]$$

$$= -17,773 / 31,982 = -0.55$$

$$\text{intercept, } a = [413 - (-0.55)272] / 7 = 80.37$$

equation of regression line is given by,

$$Y = a + bx$$

$$Y = 80.37 - 0.55x$$

Question No: 31 (Marks: 10)

The following data gives the height (in inches) of eleven 9-years old boys in a primary school.

57, 52, 51, 49, 55, 54, 50, 48, 53, 56, 47

- Find first, second and third quartiles.
- Find interquartile range, Quartile deviation.

ANS:a) Data in arranged order: 47,48,49,50,51,52,53,54,55,56,57

number of data points, n = 11

position of $Q_i = i(n+1)/4$

Position of $Q_1 = (n+1)/4 = (11+1)/4 = 12/4 = 3$

So, $Q_1 = 3^{\text{rd}}$ value

= 49

Position of $Q_2 = 2(11+1)/4 = 24/4 = 6$

So, $Q_2 = 6^{\text{th}}$ value

= 52

Position of $Q_3 = 3(11+1)/4 = 36/4 = 9$

So, $Q_3 = 9^{\text{th}}$ value

55

b) Interquartile range = $Q_3 - Q_1$

$$\begin{aligned}
 &= 55 - 49 = 6 \\
 \text{Quartile deviation} &= (Q3 - Q1) / 2 \\
 &= (55 - 49) / 2 \\
 &= 6 / 2 = 3
 \end{aligned}$$

Paper no 20
SOLVED BY CHANDA REHMAN
FINAL TERM EXAMINATION
Fall 2008

MTH302- Business Mathematics & Statistics (Session - 3)

Question No: 1 (Marks: 1) - Please choose one

6!.....

▶ **720 (Correct)**

▶ 620

▶ 420

▶ 520

Question No: 2 (Marks: 1) - Please choose one

Given : List Price = 5500Rs , Discount = 850Rs, then Net Cost Price will be
6350 Rs

▶ 5508 Rs

▶ 5585 Rs

▶ **4650 Rs (Correct)**

Question No: 3 (Marks: 1) - Please choose one

The value of x after solving the following linear equation is

$$-2x + 6 = 4x - 2$$

▶ 0

▶ 3

▶ 1/2

▶ **4/3**

Question No: 4 (Marks: 1) - Please choose one

$$A = \begin{bmatrix} 1 \end{bmatrix}$$

, then which statement is wrong about A?

- ▶ A is a column matrix.
- ▶ A is a row matrix.
- ▶ A is an identity matrix.
- ▶ **A is not a square matrix. (Correct)**

Question No: 5 (Marks: 1) - Please choose one

Percentage is a ratio of given number with standard number:

- ⌚ **100 (Correct)**
- ⌚ Same number
- ⌚ 1000
- ⌚ 10

Question No: 6 (Marks: 1) - Please choose one

Sum of annuity is always

- ▶ **Present value**
- ▶ Future value
- ▶ Net present value
- ▶ Current value

Question No: 7 (Marks: 1) - Please choose one

If the principal P is 900 rupees, time period is 6 years and rate is 12% then the simple interest will be

- ▶ **648 (Correct)**
- ▶ 658
- ▶ 668
- ▶ 678

Question No: 8 (Marks: 1) - Please choose one

In the equation $\bar{Y} = a + b\bar{X}$ the formula to calculate b is

$$b = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

▶

$$b = \frac{n \sum xy - \sum x \sum y}{\sum x^2 - (\sum x)^2}$$



$$b = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)}$$



$$b = \frac{\sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$



Question No: 9 (Marks: 1) - Please choose one

$P(A \text{ and } B) = P(A) \times P(B)$

What does this formula represent?

▶ **(a) The events A and B are Dependent. (Correct)**

▶ (b) The events A and B are Exclusive.

▶ (c) The events A and B are Independent.

▶ (d) Both (a) and (b)

Question No: 10 (Marks: 1) - Please choose one

The moving averages represent -----

▶ **Time series variations**

▶ Co-efficient of variations

▶ Statistical Dispersion

▶ Absolute deviation

Question No: 11 (Marks: 1) - Please choose one

Two dice are rolled and the numbers are added together. The probability of the total being 12 is $\frac{1}{36}$

▶ **True**

▶ **False (Correct)**

Question No: 12 (Marks: 1) - Please choose one

$$\frac{6!}{0!(6-6)!}$$

Evaluate

▶ 0

▶ 10

▶ 120

▶ 720

Question No: 13 (Marks: 1) - Please choose one

Which of the following terms is a measure of the central tendency?

▶ Standard Deviation

▶ Variance

▶ Average

▶ None of these (Correct)

Question No: 14 (Marks: 1) - Please choose one

How many arrangements can be made of the letter BUSINESS

▶ 6723

▶ 6725

▶ 6720

▶ 6721

Question No: 15 (Marks: 1) - Please choose one

There are 5 Rock songs, 6 Carnatic songs and 3 Indian pop songs. How many different albums can be formed using the above repertoire if the albums should contain at least 1 Rock song and 1 Carnatic song?

▶ 15624

▶ 16384

▶ 6144

▶ 240

REF:

Constant: 1 Rock song and 1 Carnatic song

Solution:

Rock Song has (5-1) combinations.

Hence, $2^5 - 1 = 32 - 1 = 31$ ways ----- (A)

Carnatic Song:

Carnatic Song has (6-1) combinations.

Hence, $2^6 - 1 = 64 - 1 = 63$ ways ----- (B)

Indi pop:

Indi pop can have all 3 combinations.

Hence, $2^3 = 8$ ways ----- (C)

From A, B and C we have:

$31 * 63 * 8 = 15624$ ways (Ans)

Question No: 16 (Marks: 1) - Please choose one

In an exponential smoothing model, the weighting factor w is a number that falls between _____ and _____.

▶ 0; 1

▶ -1; 1

▶ -1; 0

▶ 4; 4

▶ -2; 2

Question No: 17 (Marks: 1) - Please choose one

In a positively skewed distribution

▶ The mean, median, and mode are all equal.

▶ **The mean is larger than the median (Correct)**

▶ The median is larger than the mean.

▶ The standard deviation must be larger than the mean or the median.

Question No: 18 (Marks: 1) - Please choose one

For what purpose is the method of least squares used on time-series data?

▶ It is used for eliminating irregular movements

▶ It is used for obtaining the trend equation.

▶ It is used for deseasonalizing the data.

▶ **It is used for exponentially smoothing a series.**

Question No: 19 (Marks: 1) - Please choose one

All of the following are assumptions of the error terms in the simple linear regression model except

▶ normality

▶ error terms with a mean of zero

▶ constant variance

▶ **variance of one**

Question No: 20 (Marks: 1) - Please choose one

Which one of the following statements is not true?

▶ Correlation shows a degree of association between two variables.

► You can make any predictions with results of correlation.

- Correlation helps proceed further with regression analysis.
- You can make Predictions on the base of regression equations.

Question No: 21 (Marks: 2)

Define frequency with example.

Answer:

The number of times a certain value or class of values occur.

Question No: 22 (Marks: 2)

Define Type-I error.

Answer:

Statistical probability in hypothesis testing that the test sample supports a conclusion that a value is misstated when, in fact, the value is correctly stated. Defined also as an incorrect decision to reject something that should have been accepted, it is the mirror image of type 2 error. If the level of significance (called alpha level) for a test result is set too high, the possibility of type 1 error is reduced but the possibility of type 2 error is raised by the same factor, and vice versa. Also called alpha error or alpha risk.

Question No: 23 (Marks: 2)

A coin can be tossed in 3 ways. A die can be thrown in 6 ways. A coin and a die together can be thrown in **18**.....ways.

Question No: 24 (Marks: 3)

Determine two variable X and Y are correlated or uncorrelated by using the following table

X	0	4
Y	2	2

Question No: 25 (Marks: 3)

Show that X and Y are uncorrelated by using the following table

X	2	4
Y	3	3

Question No: 26 (Marks: 3)

Explain negative binomial distribution.

Answer:

It returns the probability that there will be number_f failures before the number_s-th success, when the constant probability of a success is probability_s. This function is similar to the binomial

distribution, except that the number of successes is fixed, and the number of trials is variable. Like the binomial, trials are assumed independent.

Question No: 27 (Marks: 5)

What will be the

$$\sum (X - \bar{X})(Y - \bar{Y}) = 5.$$

Question No: 28 (Marks: 5)

On a final examination in mathematics the mean was 72 and the standard deviation was 15. Determine the standard scores of students receiving marks
(a) 60, (b) 93, (c) 72

Question No: 29 (Marks: 5)

a) Write all the combinations of $abcd$ taken 1 at a time.

Answer : $4!/1!(4-1)! = 4$

b) Write their combinations taken 2 at a time.

Answer : $4!/2!(4-2)! = 4.3.2/2.2 = 6$

c) Write their combinations taken 3 at a time.

Answer : $4!/3!(4-3)! = 4.3.2/3.2 = 6$

d) Write their combinations taken 4 at a time.

Answer : $4!/4!(4-4)! = 1$

Question No: 30 (Marks: 10)

Find the standard deviation and variance for 10 randomly selected data values :

44, 50, 38, 96, 42, 47, 40, 39, 46, 50 .

Answer:

Standard Deviation = 19.99136

Question No: 31 (Marks: 10)

Find the mean, median, mode, and range for the following list of values:
13, 18, 13, 14, 13, 16, 14, 21, 13

Answer:

Mean = 15 (N=9)

Median = 14

Mode = 13

Range = 13